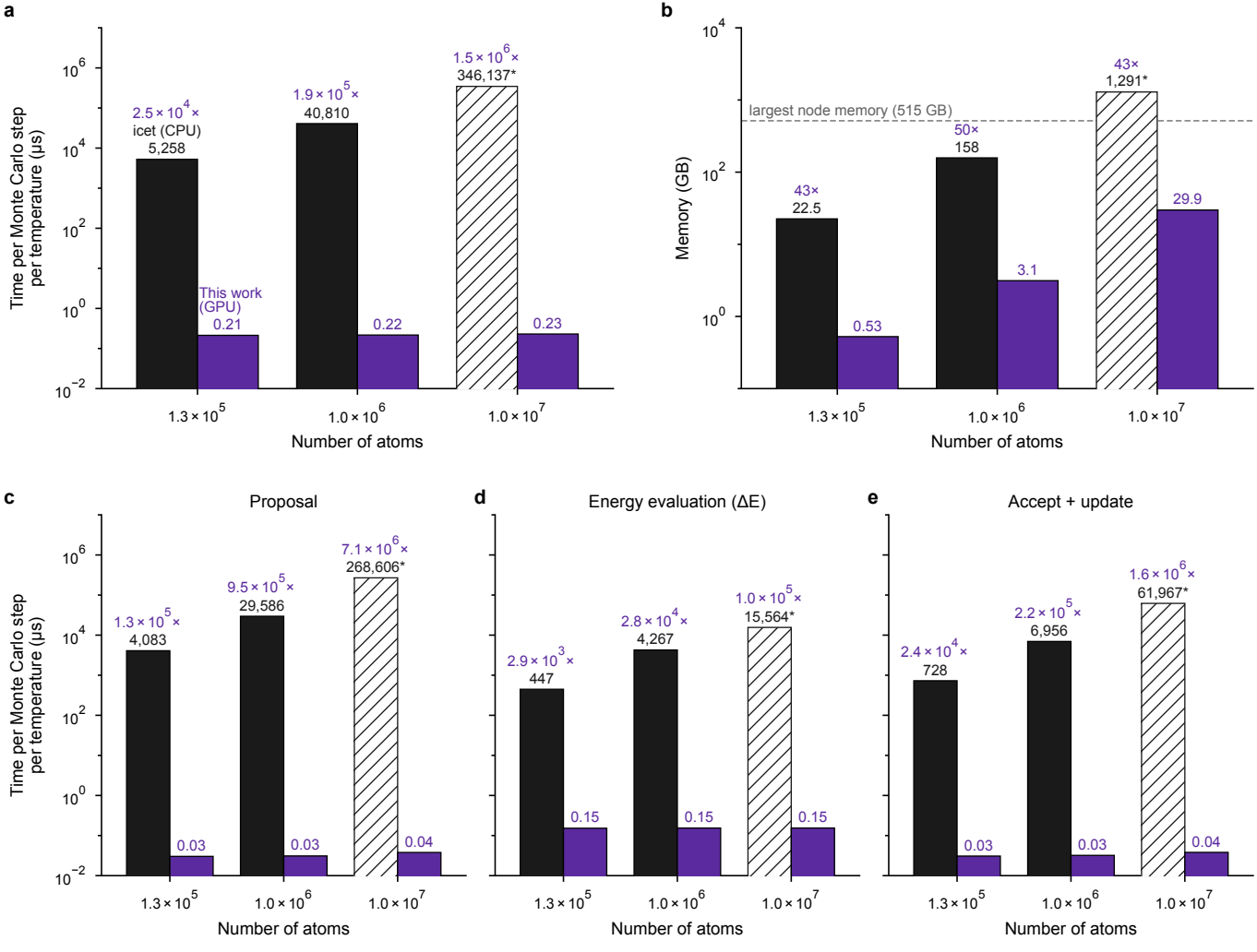
a
Time per Monte Carlo step per temperature (μs)
10^6
10^4
10^2
10^0
10^{-2}
$2.5 \times 10^4\times$
icet (CPU)
5,258
This work (GPU)
0.21
$1.9 \times 10^5\times$
40,810
0.22
$1.5 \times 10^6\times$
346,137*
0.23
1.3×10^5
1.0×10^6
1.0×10^7
Number of atoms
b
Memory (GB)
10^4
10^2
10^0
largest node memory (515 GB)
43×
22.5
0.53
50×
158
3.1
43×
1,291*
29.9
1.3×10^5
1.0×10^6
1.0×10^7
Number of atoms
c
Proposal
Time per Monte Carlo step per temperature (μs)
10^6
10^4
10^2
10^0
10^{-2}
$1.3 \times 10^5\times$
4,083
0.03
$9.5 \times 10^5\times$
29,586
0.03
$7.1 \times 10^6\times$
268,606*
0.04
1.3×10^5
1.0×10^6
1.0×10^7
Number of atoms
d
Energy evaluation (ΔE)
$2.9 \times 10^3\times$
447
0.15
$2.8 \times 10^4\times$
4,267
0.15
$1.0 \times 10^5\times$
15,564*
0.15
1.3×10^5
1.0×10^6
1.0×10^7
Number of atoms
e
Accept + update
$2.4 \times 10^4\times$
728
0.03
$2.2 \times 10^5\times$
6,956
0.03
$1.6 \times 10^6\times$
61,967*
0.04
1.3×10^5
1.0×10^6
1.0×10^7
Number of atoms